Sdn project

Sampreeth Jaladi

PES1UG24AM246

Ss1:

```
samp@Sampreeth:/mnt/c/Users/sampr$ cd "/mnt/C:/CN PROJECT"
-bash: cd: /mnt/C:/CN PROJECT: No such file or directory
samp@Sampreeth:/mnt/c/Users/sampr$ cp "/mnt/c/CN PROJECT/static_routing_controlle
r.py" ~/pox/ext/
cd ~/pox
python3 pox.py log.level --DEBUG static_routing_controller
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:static_routing_controller:========================================
INFO:static_routing_controller: Static Routing Controller Initialized
INFO:static_routing_controller:========================================
INFO:static_routing_controller:Static Routing Controller module loaded
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 3 2026 12:15:18)
DEBUG:core:Platform is Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.
39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8
 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:static_routing_controller:----------------------------------------
INFO:static_routing_controller:Switch s1 connected (dpid=00-00-00-00-00-01)
INFO:static_routing_controller:  [s1] IP Route:  dst=10.0.0.1 -> port 1 (mac=00:0
0:00:00:00:01)
INFO:static_routing_controller:  [s1] IP Route:  dst=10.0.0.2 -> port 2 (mac=00:0
0:00:00:00:02)
INFO:static_routing_controller:  [s1] IP Route:  dst=10.0.0.3 -> port 3 (mac=00:0
0:00:00:00:03)
INFO:static_routing_controller:  [s1] IP Route:  dst=10.0.0.4 -> port 3 (mac=00:0
0:00:00:00:04)
INFO:static_routing_controller:  [s1] ARP Rule: dst=10.0.0.1 -> port 1
INFO:static_routing_controller:  [s1] ARP Rule: dst=10.0.0.2 -> port 2
INFO:static_routing_controller:  [s1] ARP Rule: dst=10.0.0.3 -> port 3
INFO:static_routing_controller:  [s1] ARP Rule: dst=10.0.0.4 -> port 3
INFO:static_routing_controller:  [s1] Table-miss rule installed
INFO:static_routing_controller:  [s1] All 9 flow rules installed successfully
INFO:static_routing_controller:----------------------------------------
```

Ss2:

```
╔═══════════════════════════════════════════════════════╗
║         STATIC ROUTING SDN - NETWORK RUNNING          ╒
╔═══════════════════════════════════════════════════════╗
║                                                       ╒
  h1(.0.1) ─── s1 ──────── s2 ─── h2(.0.2)             ╒
                │           │                           ╒
               s3 ────────  s4                          ╒
                │           │                           ╒
  h3(.0.3)     '  h4(.0.4) '                            ╒
╔═══════════════════════════════════════════════════════╗
║  TEST COMMANDS:                                       ╒
║    pingall                    All-to-all ping         ╒
║    h1 ping -c 5 h2           Direct path test         ╒
║    h1 ping -c 5 h4           Multi-hop path test      ╒
║    iperf h1 h2               Throughput (direct)      ╒
║    iperf h1 h4               Throughput (multi-hop)   ╒
║    sh ovs-ofctl dump-flows s1   View flow table       ╒
╔═══════════════════════════════════════════════════════╗

*** Starting CLI:
mininet> |
```

Ss3:

```
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> |
```

Ss 4 :

```
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=12.2 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.5 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.9 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 10.820/11.248/12.187/0.544 ms
mininet> |
```

Ss 5 :

```
mininet> h1 ping -c 5 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=23.5 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=23.3 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=23.0 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=22.5 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=23.5 ms

--- 10.0.0.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 22.485/23.166/23.540/0.399 ms
mininet>
```

Ss6:

```
mininet> h1 iperf -c 10.0.0.2 -t 5
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
tcp connect failed: Connection refused
[  1] local 0.0.0.0 port 0 connected with 10.0.0.2 port 5001
mininet> mininet> h2 iperf -s &
*** Unknown command: mininet> h2 iperf -s &
mininet> h2 iperf -s &
mininet> h1 iperf -c 10.0.0.2 -t 5
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 45228 connected with 10.0.0.2 port 5001 (icwnd/mss/irtt
=14/1448/12425)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-7.4059 sec  11.0 MBytes  12.5 Mbits/sec
mininet>
```

Ss7:

```
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=144.717s, table=0, n_packets=6433, n_bytes=424634, priority
=100,ip,nw_dst=10.0.0.1 actions=mod_dl_dst:00:00:00:00:00:01,output:"s1-eth1"
 cookie=0x0, duration=144.672s, table=0, n_packets=6730, n_bytes=19580852, priori
ty=100,ip,nw_dst=10.0.0.2 actions=mod_dl_dst:00:00:00:00:00:02,output:"s1-eth2"
 cookie=0x0, duration=144.672s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.3 actions=mod_dl_dst:00:00:00:00:00:03,output:"s1-eth3"
 cookie=0x0, duration=144.672s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.4 actions=mod_dl_dst:00:00:00:00:00:04,output:"s1-eth3"
 cookie=0x0, duration=144.672s, table=0, n_packets=4, n_bytes=168, priority=100,a
rp,arp_tpa=10.0.0.1 actions=output:"s1-eth1"
 cookie=0x0, duration=144.672s, table=0, n_packets=4, n_bytes=168, priority=100,a
rp,arp_tpa=10.0.0.2 actions=output:"s1-eth2"
 cookie=0x0, duration=144.672s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.3 actions=output:"s1-eth3"
 cookie=0x0, duration=144.672s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.4 actions=output:"s1-eth3"
 cookie=0x0, duration=144.672s, table=0, n_packets=3, n_bytes=210, priority=0 act
ions=CONTROLLER:65535
```

Ss 8 :

```
mininet> sh ovs-ofctl dump-flows s3
 cookie=0x0, duration=175.113s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.1 actions=mod_dl_dst:00:00:00:00:00:01,output:"s3-eth1"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.2 actions=mod_dl_dst:00:00:00:00:00:02,output:"s3-eth1"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.3 actions=mod_dl_dst:00:00:00:00:00:03,output:"s3-eth2"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,ip,
nw_dst=10.0.0.4 actions=mod_dl_dst:00:00:00:00:00:04,output:"s3-eth3"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.1 actions=output:"s3-eth1"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.2 actions=output:"s3-eth1"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.3 actions=output:"s3-eth2"
 cookie=0x0, duration=175.072s, table=0, n_packets=0, n_bytes=0, priority=100,arp
,arp_tpa=10.0.0.4 actions=output:"s3-eth3"
```

Ss9 :

```
mininet> h1 ping -c 10 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=24.4 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=22.1 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=20.6 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=20.6 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=22.4 ms
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=22.3 ms
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=22.9 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=22.9 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=22.4 ms
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=22.2 ms

--- 10.0.0.4 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 8996ms
rtt min/avg/max/mdev = 20.608/22.274/24.363/1.031 ms
mininet> sh ovs-ofctl dump-flows s3
 cookie=0x0, duration=244.539s, table=0, n_packets=10, n_bytes=980, priorit
y=100,ip,nw_dst=10.0.0.1 actions=mod_dl_dst:00:00:00:00:00:01,output:"s3-et
h1"
 cookie=0x0, duration=244.498s, table=0, n_packets=0, n_bytes=0, priority=1
00,ip,nw_dst=10.0.0.2 actions=mod_dl_dst:00:00:00:00:00:02,output:"s3-eth1"
 cookie=0x0, duration=244.498s, table=0, n_packets=0, n_bytes=0, priority=1
00,ip,nw_dst=10.0.0.3 actions=mod_dl_dst:00:00:00:00:00:03,output:"s3-eth2"
 cookie=0x0, duration=244.498s, table=0, n_packets=10, n_bytes=980, priorit
y=100,ip,nw_dst=10.0.0.4 actions=mod_dl_dst:00:00:00:00:00:04,output:"s3-et
h3"
 cookie=0x0, duration=244.498s, table=0, n_packets=2, n_bytes=84, priority=
100,arp,arp_tpa=10.0.0.1 actions=output:"s3-eth1"
 cookie=0x0, duration=244.498s, table=0, n_packets=0, n_bytes=0, priority=1
00,arp,arp_tpa=10.0.0.2 actions=output:"s3-eth1"
 cookie=0x0, duration=244.498s, table=0, n_packets=0, n_bytes=0, priority=1
00,arp,arp_tpa=10.0.0.3 actions=output:"s3-eth2"
 cookie=0x0, duration=244.498s, table=0, n_packets=2, n_bytes=84, priority=
100,arp,arp_tpa=10.0.0.4 actions=output:"s3-eth3"
 cookie=0x0, duration=244.498s, table=0, n_packets=4, n_bytes=280, priority
=0 actions=CONTROLLER:65535
mininet>
```

Ss 10 :

```
mininet>  sh ovs-ofctl del-flows s1
mininet>  sh ovs-ofctl del-flows s2
mininet>  sh ovs-ofctl del-flows s3
mininet>  sh ovs-ofctl del-flows s4
mininet> sh ovs-ofctl dump-flows s1
mininet> sh ovs-ofctl dump-flows s1
mininet> sh ovs-ofctl dump-flows s1
```

Ss11:

```
mininet>  h1 ping -c 3 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.

--- 10.0.0.4 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2006ms

mininet>
```

Ss12 :

```
mininet> sh ovs-vsctl del-controller s1
mininet> sh ovs-vsctl set-controller s1 tcp:127.0.0.1:6633
mininet> sh ovs-vsctl del-controller s2
mininet>  sh ovs-vsctl set-controller s2 tcp:127.0.0.1:6633
mininet> sh ovs-vsctl del-controller s3
mininet> sh ovs-vsctl set-controller s3 tcp:127.0.0.1:6633
mininet> sh ovs-vsctl del-controller s4
mininet> sh ovs-vsctl set-controller s4 tcp:127.0.0.1:6633
mininet> h1 ping -c 3 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=47.8 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=22.5 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=22.4 ms

--- 10.0.0.4 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 22.408/30.916/47.793/11.933 ms
mininet>
```